

SCHEME OF VALUATION

CST201 DATA STRUCTURES · Sample paper · Max. Marks: 23

PART A (3 × 3 = 9 marks)

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|----------|---|----------|
| 1 | Time complexity: the amount of time an algorithm takes as a function of input size n (1). Big-O: an upper bound on the growth rate, $f(n) = O(g(n))$ if $f(n) \leq c \cdot g(n)$ for $n \geq n_0$ (1). Any correct example, e.g. linear search is $O(n)$ (1). | 3 |
| 2 | Stack is LIFO (1). Queue is FIFO (1). Stack inserts/deletes at one end (push/pop, top); queue inserts at rear and deletes at front (enqueue/dequeue) (1). | 3 |
| 3 | Correct postfix: $A B C * +$ (3). Wrong operator precedence (e.g. $A B + C *$): 0, but award 1 if the operator-stack method is shown correctly. | 3 |

PART B (any one full question, 14 marks)

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|------------|---|----------|
| 4 a | Create a new node / allocate memory (2); store the data in the new node (1); new node's next = head (2); head = new node (2). | 7 |
| 4 b | Definition: a queue whose last position is connected back to the first, forming a circle (2); neat labelled diagram showing front and rear (2); insertion/deletion using $\text{rear} = (\text{rear} + 1) \% \text{size}$ and $\text{front} = (\text{front} + 1) \% \text{size}$ (2); advantage over a linear queue: space freed at the front is reused, no wasted space (1). | 7 |
| 5 a | Works on a sorted array (1); compare with the middle element and search the left or right half (3); correct worked example (2); complexity $O(\log n)$ (1). | 7 |
| 5 b | Correct bubble sort algorithm with nested loops and swap (4); time complexity $O(n^2)$ worst/average (2), $O(n)$ best with a swap flag or equivalent (1). | 7 |

If a student answers both Q4 and Q5, count the better one. Award marks for correct points even if worded differently.